

Diabetes Prediction using Machine Learning

Logistic Regression, Random Forest, and SVM Benchmarking

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Abstract

Early prediction of diabetes can significantly improve treatment outcomes and patient quality of life. In this project, we developed a machine learning pipeline to predict diabetes using patient health indicators such as glucose level, BMI, age, and insulin measurements. We benchmarked three models — Logistic Regression, Random Forest, and Support Vector Machine (SVM) — using GridSearchCV for hyperparameter tuning. Performance was evaluated on accuracy, precision, recall, F1 score, and ROC AUC. The Random Forest achieved the best classification accuracy and F1 score, while SVM obtained the highest ROC AUC, highlighting different strengths depending on the evaluation criterion.

1 Introduction

Diabetes is a chronic disease that requires early detection and intervention. Machine learning offers a powerful approach for predictive diagnostics by leveraging patient health data. This project explores three supervised learning models applied to the Diabetes dataset. Our objective is to compare their performance, analyze trade-offs, and identify the most suitable model for different medical use cases.

2 Dataset and Preprocessing

The dataset consists of 768 patient records with 8 clinical features and a binary outcome:

- Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, Age, Pregnancies
- Outcome: 0 = Non-diabetic, 1 = Diabetic

Data Cleaning

- Replaced medically invalid zeros in Glucose, BloodPressure, SkinThickness, Insulin, BMI with median values.
- Standardized features using `StandardScaler`.
- Applied an 80-20 train-test split with stratification to preserve class balance.

3 Exploratory Data Analysis (EDA)

Key insights:

- The dataset is moderately imbalanced (65% non-diabetic vs 35% diabetic).

- Higher glucose and BMI values strongly correlate with diabetic outcomes.
- Age distribution indicates increased diabetes risk in older patients.

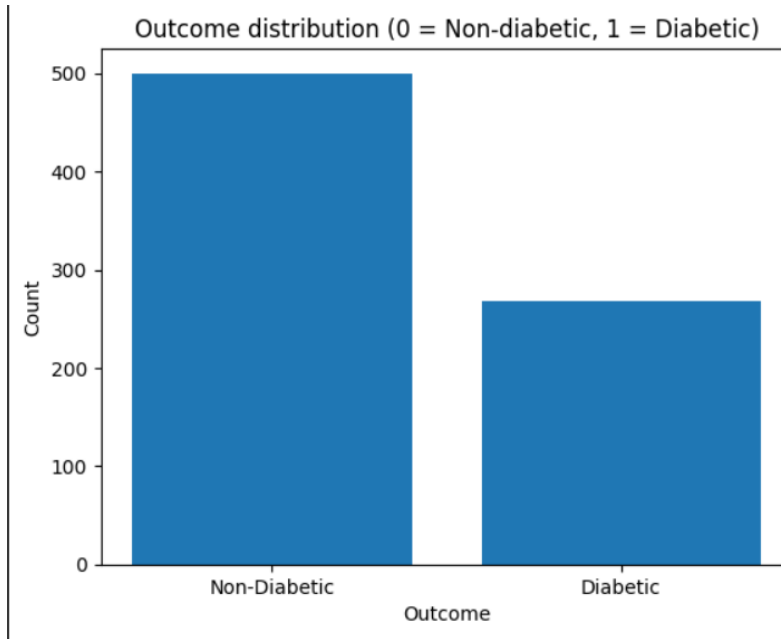


Figure 1: Outcome distribution in the dataset.

4 Methodology

4.1 Models Implemented

1. **Logistic Regression** — interpretable baseline, linear decision boundaries.
2. **Random Forest** — ensemble of decision trees, captures nonlinear interactions, tuned for `max_depth`, `min_samples_split`, and `n_estimators`.
3. **Support Vector Machine (SVM)** — RBF kernel, tuned for `C` and `gamma` with Grid-SearchCV.

5 Results

The models were evaluated on test data using accuracy, precision, recall, F1 score, and ROC AUC.

Model	Accuracy	Precision	Recall	F1	ROC AUC
Logistic Regression	0.71	0.60	0.50	0.55	0.813
Random Forest	0.74	0.65	0.57	0.61	0.808
SVM (RBF)	0.69	0.58	0.48	0.53	0.817

Table 1: Model performance comparison.

Key Observations

- **Logistic Regression:** Competitive ROC AUC but lower recall (misses more diabetics).

- **Random Forest:** Best overall accuracy and F1, robust to feature interactions.
- **SVM:** Highest ROC AUC, meaning excellent ranking ability, but weaker accuracy/F1 without calibration.

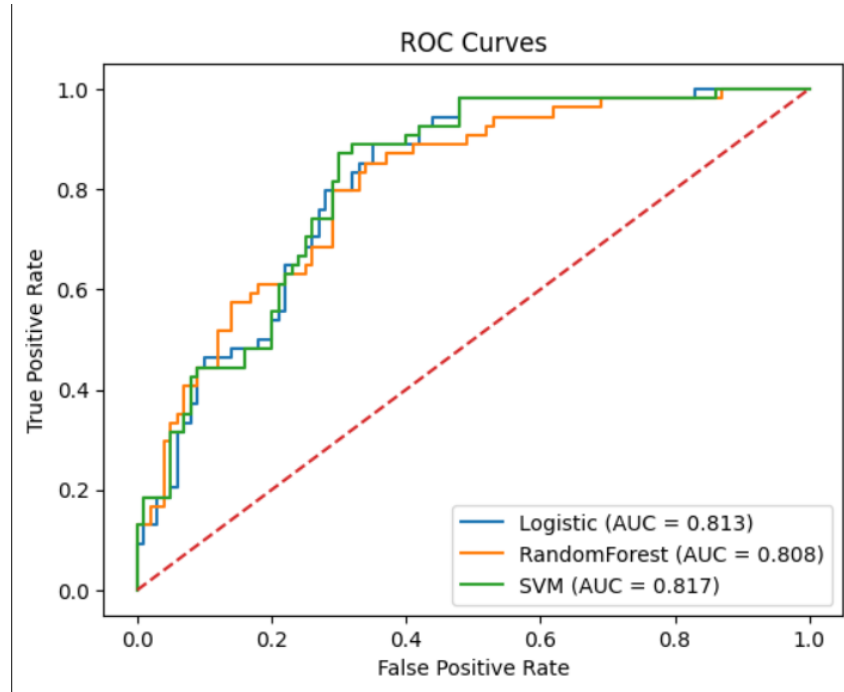


Figure 2: ROC curves for Logistic Regression, Random Forest, and SVM.

6 Discussion

Interpretation

- Random Forest is the most practical classifier when both accuracy and recall are important.
- SVM is valuable when ranking patients by diabetes risk (e.g., for screening), but needs probability calibration.
- Logistic Regression remains useful for interpretability and clinical transparency.

7 Conclusion

This project demonstrates the importance of benchmarking multiple models for medical prediction tasks. While Random Forest achieved the best F1 score, SVM provided the best ROC AUC. The choice of model depends on the clinical objective: minimizing false negatives (recall) or achieving robust ranking of patients (ROC AUC).

Future Work

- Apply **CalibratedClassifierCV** to SVM for better probability estimates.
- Experiment with advanced models such as XGBoost and LightGBM.

- Perform threshold tuning to prioritize recall in real-world screening.
- Use cross-validation with nested hyperparameter tuning for robustness.

References

1. Scikit-learn Documentation: <https://scikit-learn.org>